# 120 Days of Quantitative Finance

Month 5: Advanced Quant Systems

Week 17

Day 116: Implied Volatility and Volatility Surface

## 1 Introduction

Implied volatility is the volatility parameter that matches the observed market price of an option.

It is obtained by inverting the Black-Scholes formula.

## 2 Definition

$$C_{market} = C_{BS}(S, K, T, r, \sigma_{imp})$$

## 3 Numerical Solution

There is no closed-form solution for $\sigma_{imp}$.

We solve:

$$f(\sigma) = C_{BS}(\sigma) - C_{market} = 0$$

using numerical methods:

- Newton-Raphson
- Bisection method

## 4 Newton-Raphson Update

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{market}}{Vega}$$

## 5 Volatility Smile

For fixed maturity:

$$\sigma_{imp} = f(K)$$

## 6 Volatility Surface

Across strikes and maturities:

$$\sigma_{imp} = f(K, T)$$

## 7 Financial Interpretation

- Reflects market expectations
- Includes risk premium
- Captures supply-demand dynamics

## 8 Applications

- option pricing
- model calibration
- volatility trading

## 9 Conclusion

Implied volatility is a central concept in derivatives markets, linking observed prices to model parameters.